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# Complements of Operational Research

## Exercise sheet 8 — Multi-criteria decision making (AHP)

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The exercises marked **homework** are to be solved at home and sent by email to the lecturer, by the deadline given in class; the solution is shared afterwards. The last one is from a previous test. Unless stated otherwise, the weights are obtained by column normalisation; consider a consistency ratio  $CR < 0.10$  acceptable ( $RI_3 = 0.58$ ,  $RI_4 = 0.90$ ,  $RI_5 = 1.12$ ).

### Exercise 1 — AHP with 3 criteria and 3 alternatives

Homework (Moodle, Python solution)

We want to choose the best of three alternatives, A, B and C, based on three criteria, C1, C2 and C3.

The criteria comparison matrix is  $\begin{pmatrix} 1 & 1/2 & 4 \\ 2 & 1 & 7 \\ 1/4 & 1/7 & 1 \end{pmatrix}$  and the alternatives comparison matrices for C1, C2 and C3 are

$$\begin{pmatrix} 1 & 1/3 & 3 \\ 3 & 1 & 5 \\ 1/3 & 1/5 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 5 & 1/7 \\ 1/5 & 1 & 1/8 \\ 7 & 8 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 1/3 \\ 1/2 & 1 & 1/4 \\ 3 & 4 & 1 \end{pmatrix}.$$

- Compute the criteria weights and the consistency ratio of their matrix.
- Compute the priorities of the alternatives on each criterion and the corresponding consistency ratios.
- Compute the final score and state the chosen alternative. Comment on the consistency of the data.

### Exercise 2 — AHP with 3 criteria and 5 alternatives

Homework (Moodle, Python solution)

Same criteria matrix as in Exercise 1, now with five alternatives, A to E. Alternatives comparison matrices for C1, C2 and C3:

$$\begin{pmatrix} 1 & 1/3 & 3 & 1/2 & 2 \\ 3 & 1 & 5 & 1/3 & 1/4 \\ 1/3 & 1/5 & 1 & 1/7 & 1/2 \\ 2 & 3 & 7 & 1 & 4 \\ 1/2 & 4 & 2 & 1/4 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 5 & 1/7 & 3 & 1/3 \\ 1/5 & 1 & 1/8 & 1/4 & 1/5 \\ 7 & 8 & 1 & 5 & 2 \\ 1/3 & 4 & 1/5 & 1 & 1/2 \\ 3 & 5 & 1/2 & 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 1/3 & 4 & 1/2 \\ 1/2 & 1 & 1/4 & 3 & 1/3 \\ 3 & 4 & 1 & 5 & 2 \\ 1/4 & 1/3 & 1/5 & 1 & 1/7 \\ 2 & 3 & 1/2 & 7 & 1 \end{pmatrix}.$$

- Compute the priorities of the alternatives on each criterion and the consistency ratios.
- Compute the final score and rank the alternatives.
- Should any of the matrices be sent back to the decision maker? Why?

### Exercise 3 — Supplier selection with verbal judgements

Homework (Moodle, MATLAB solution)

Company XYZ assesses two suppliers, A and B, based on cost, quality and delivery time. Cost is considered twice as important as quality and three times as important as delivery time; quality is four times as important as delivery time. On cost, supplier B is 3 times preferable to A; on quality, A is 5 times preferable to B; on delivery time, B is 7 times preferable to A.

- Write the comparison matrices.
- Compute the criteria weights and the consistency ratio.
- Which supplier should be chosen?

**Exercise 4 — AHP with 2 criteria and 5 alternatives**

Homework (Moodle, MATLAB solution)

An organisation is choosing between five alternatives, 1 to 5, with two criteria; criterion 1 is considered twice as important as criterion 2. Alternatives comparison matrices for criteria 1 and 2:

$$\begin{pmatrix} 1 & 3 & 7 & 1/2 & 1/3 \\ 1/3 & 1 & 5 & 1/5 & 1/7 \\ 1/7 & 1/5 & 1 & 1/9 & 1/8 \\ 2 & 5 & 9 & 1 & 1/2 \\ 3 & 7 & 8 & 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1/3 & 1/5 & 3 & 3 \\ 3 & 1 & 1/3 & 5 & 5 \\ 5 & 3 & 1 & 6 & 6 \\ 1/3 & 1/5 & 1/6 & 1 & 1 \\ 1/3 & 1/5 & 1/6 & 1 & 1 \end{pmatrix}.$$

- Compute the priorities of the alternatives on each criterion and check the consistency.
- Find the final score and the best alternative.
- Why does the criteria matrix not need a consistency check?

**Exercise 5 — AHP by the eigenvector method**

Homework (Moodle, Python solution)

We want to choose the best of five alternatives, A to E, with three criteria: cost, quality and time.

Criteria matrix:  $\begin{pmatrix} 1 & 2 & 1/2 \\ 1/2 & 1 & 1/3 \\ 2 & 3 & 1 \end{pmatrix}$ . Alternatives matrices for cost, quality and time:

$$\begin{pmatrix} 1 & 1/3 & 5 & 2 & 1/4 \\ 3 & 1 & 7 & 3 & 1/2 \\ 1/5 & 1/7 & 1 & 1/3 & 1/8 \\ 1/2 & 1/3 & 3 & 1 & 1/3 \\ 4 & 2 & 8 & 3 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 4 & 1/2 & 3 & 7 \\ 1/4 & 1 & 1/5 & 1/2 & 3 \\ 2 & 5 & 1 & 4 & 9 \\ 1/3 & 2 & 1/4 & 1 & 5 \\ 1/7 & 1/3 & 1/9 & 1/5 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 & 1/2 & 2 & 2 \\ 1/3 & 1 & 1/4 & 1/2 & 1/2 \\ 2 & 4 & 1 & 3 & 3 \\ 1/2 & 2 & 1/3 & 1 & 1 \\ 1/2 & 2 & 1/3 & 1 & 1 \end{pmatrix}.$$

- Using `numpy.linalg.eig`, obtain the priority vectors from the eigenvector associated with the largest eigenvalue of each matrix.
- Compute the consistency indices and ratios.
- Aggregate the results and choose the best alternative.
- Repeat with column normalisation. Does the choice change?

**Exercise 6 — Location of a health centre**

Homework (new)

A town council is choosing between three sites, L1, L2 and L3, for a new health centre, with three criteria: population served within 15 minutes (L1: 42 000, L2: 35 000, L3: 28 000 inhabitants), cost of the land and of the construction (L1: 3.2 M€, L2: 2.6 M€, L3: 2.1 M€) and accessibility by public transport. On accessibility, L2 is moderately preferable to L1 (3), L1 is slightly preferable to L3 (2) and L2 is strongly preferable to L3 (5). Population is moderately more important than cost (3) and slightly more important than accessibility (2); accessibility is slightly more important than cost (2).

- Compute the criteria weights and the consistency ratios.
- Compute the local priorities (population as a direct attribute, cost as indirect) and state the chosen site.
- The councillor for finance wants to give more weight to cost, keeping the proportions between the other two criteria. From what weight of cost does the choice change? Which site would then be chosen?
- Below what weight of accessibility does the choice change? Comment on the robustness of the decision.

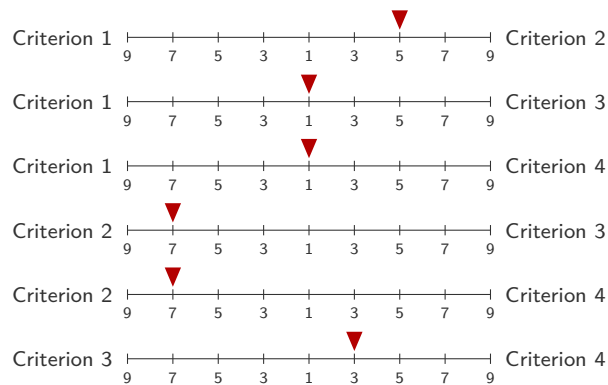
### Exercise 7 — Supplier selection

2nd test, 11/06/2024 (part c new)

A company has to decide which supplier to select, considering the following criteria: 1st the raw-material cost, 2nd the transport cost, 3rd the procurement lead time and 4th the raw-material yield per tonne. The values offered by each supplier are given in the following table:

|            | RM cost (ton.) | Transport cost | Procurement lead time | RM yield per ton. |
|------------|----------------|----------------|-----------------------|-------------------|
| Supplier 1 | 120            | 1500           | 15                    | 2.5               |
| Supplier 2 | 150            | 1000           | 12                    | 4.5               |
| Supplier 3 | 100            | 700            | 14                    | 3.5               |
| Supplier 4 | 130            | 1300           | 16                    | 4                 |

The company's management department met and set the following preferences when comparing the criteria, according to Saaty's scale:



The OR department met and, based on the information gathered and using the usual ranking of direct and indirect attributes, presented an answer.

- Are the data given by the managers consistent?
- Which supplier was selected? Justify.
- Which criterion's weight can change the choice most easily, and from what value?